

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT OF COMPUTER AND SOFTWARE ENGINEERING

Year I

WEB TECHNOLOGY

Prepared by: ICT School-CST-UR

UNIT 1:

Content.

- **Introduction to the Internet**
- **History and overview of the Web**

1. Introduction to the Internet

The Internet has been called the "network of networks." Simply put, it's a way to link millions of people around the world, each of whom is using a computer connected to a smaller network such as the one in their office, or school, or to a commercial network such as America Online or CompuServe.

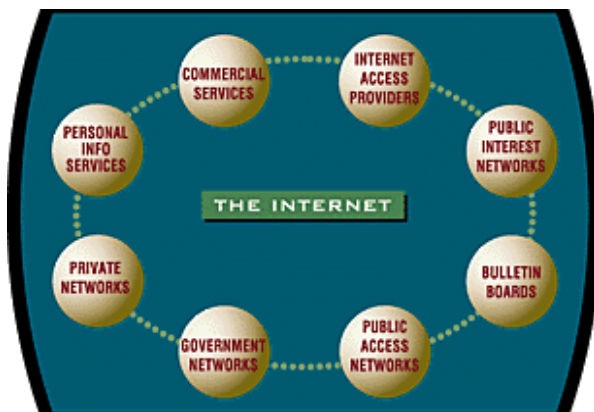


Figure 1-1: Structure of the interconnected things (IoT)

The Internet has quickly become a global communications system, offering new ways to reach other people and vast new sources of information. Don't be overwhelmed by the rapid advance of the technology. The phone lines and computers are merely the tools of the networked world that make new forms of human communication possible. These new ways to communicate are supported by a variety of simple tools for using the Internet. One aspect of the Internet, the World Wide Web, is becoming one of the most popular forms of this new communications medium. Nobody owns or controls the Internet or the World Wide Web, although millions of individuals and organizations control their own piece of it. The Internet does have some governing bodies that propose standards and specifications and help plan for the future including the Internet Society and the World Wide Web Consortium and the U.S. Internet Council.

Individuals or organizations who just want to browse the Internet, send e-mail or use other basic facilities will need only a computer, modem, communications software and a simple way of accessing the Internet. The faster the computer and modem, the better, especially if you want to use the most sophisticated graphics and the animation capabilities of the World Wide Web. However, even older, slower computers can make use of most of the Internet's information and

communications resources. It doesn't have to cost much to become an accomplished network user. Even those who want to create a World Wide Web presence will find many affordable options, although ambitious Web sites will require more sophisticated equipment and services. The explosive growth of the Internet and the World Wide Web has created an environment where users have many choices and few restrictions. The sheer number of resources can be overwhelming, but there are a number of aids for navigating the Internet that can help you find the information you need.

1.1: What is the Internet?

The Internet is the GSI (Global System of Interconnected) network of computers, which is connected with different types of computers whole over the world. In the word, "INTERNET" was coming from "Inter-Networking," a mechanism of the Internet to transfer data from a computer to the other computers.

The Internet means an Interconnection of Networks, which are communication protocols for communications between two computers over the network. The Internet is a huge capability of stored data such as music, video, software and any types of documents. The Internet is an electronic web which is connected to people each other's that have access to networks allows them to send and receive E-mails and participate in a number of other activities. In the present situation 80% of people access, the Internet and lots of people busy to finding music, video and software and huge people visiting the social networking site like Facebook, Twitter, Myspace, linked In, etc. Today the Internet is called by "NET" then the word "net" is most popular of every human life.



Figure 1-2: Internet

1.2: History of the Internet

The Internet is one of the most successful technologies in the world. Now the Internet enormously used development information infrastructure and commitment to research. In 1966, develop the computer network idea a plan for the ARPANET and quickly put together and publish it 1967. The ARPANET system comes from Advanced Research Project Agency (ARPA) of the U.S. Department of defense. ARPANET was the first WAN which is only four sites in 1969. The Internet evolved from the basic ideas of the ARPANET for interconnecting computers. At first, research organizations and used it to share and exchange information. In 1989, the U.S. Government lifted controls on the use of the Internet, and allowed its use for commercial purposes as well. Since then, the Internet has grown rapidly to become the world's largest network. It now interconnects more than 50,000 networks, allowing more than 15 million computers, more than 100 million computer users and more than 155 countries around the world easily communicate with each other. The Internet network continues to grow at the rapid pace.



ARPANET computer

Figure 1-3: ARPANET Computer

1.3: Use of the INTERNET

Internet is today one of the most important part of our daily life. There are large numbers of things that can be done using the internet and so it is very important. You can say that with the

progress in the internet we are progressing in every sphere of life as it not only makes our tasks easier but also saves a lot of time. Today internet is used for different purposes depending upon the requirement. Here in this very article we have mentioned then ten best uses of the internet. Here goes the list.

- Communication
- Research
- Education
- Financial Transaction
- Real Time Updates
- Online Booking
- Job Search
- Shopping
-

1.4: Internet tools

The Internet has many more tools but here are the 4 most popular tools, to make efficient use of the net.

- Information retrieval tools.
- Communication tools.
- Multimedia information tools.
- Information search tools.

1.5: Who Pays for the Internet?

There is no clear answer to this question because the Internet is not one "thing", it's many things. No one central agency exists that charges individual Internet users. Rather, individuals and institutions who use the Internet pay a local or regional Internet service provider for their share of services. And in turn, those smaller Internet service providers might purchase services from an even larger network. So basically, everyone who uses the Internet in some way pays for part of it.

1.6: What makes the internet work?

The unique thing about the Internet is that it allows many different computers to connect and talk to each other. This is possible because of a set of standards, known as **protocols** that govern the transmission of data over the network: TCP/IP (Transmission Control Protocol/Internet Protocol). Most people who use the Internet aren't so interested in details related to these protocols. They do, however, want to know what they can do on the Internet and how to do it effectively.

1.7: The Internet Properties

Key properties of the Internet:

- The Internet is interoperable.
- The Internet is global.
- The Web makes it easy.
- The costs of the network are shared across multiple applications and borne by the end users.

The striking characteristic of the Internet --> heterogeneity.

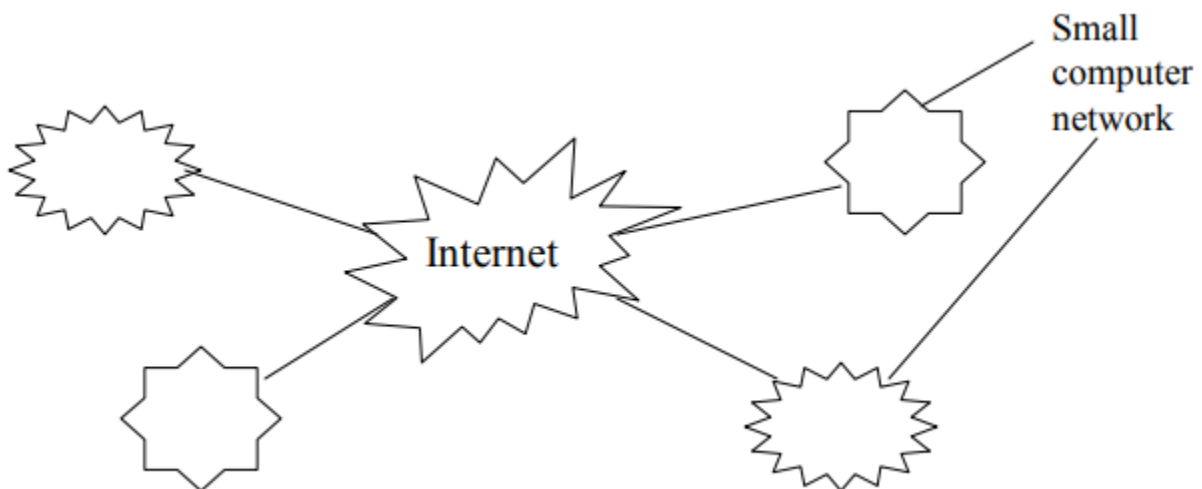


Figure 1-5: Internet Properties

2. History and Overview of the Web

2.1: Overview of WWW

Although many people use the terms World Wide Web and Internet interchangeably, the World Wide Web actually is a service of the Internet. While the Internet was developed in the late 1960s, the World Wide Web emerged in the early 1990s. Since then, it has grown phenomenally to become one of the more widely used Internet services. The **World Wide Web (WWW)**, or **Web**, consists of a worldwide collection of electronic documents and services, distributed across the Internet and linked together by hypertext links. Each electronic document on the Web is called a **Web page**, which can contain text, graphics, animation, audio, and video. Additionally, Web pages usually have built-in connections to other documents.

Some Web pages are static (fixed); others are dynamic (changing). Visitors to a static Web page all see the same content. With a dynamic Web page, by contrast, visitors can customize some or all of the viewed content such as desired stock quotes, weather for a region, or ticket availability for flights. A **Web site** is a collection of related Web pages and associated items, such as documents and pictures, stored on a Web server. A **Web server** is a computer that delivers requested Web pages to your computer. The same Web server can store multiple Web sites. Some industry experts use the term **Web 2.0** to refer to Web sites that provide a means for users to share personal information (such as social networking Web sites), allow users to modify Web site content (such as wikis), and have application software built into the site for visitors to use (such as e-mail and word processing programs). Read Looking Ahead 2-1 for a look at Web 3.0.

2.2: Evolution of WWW

World Wide Web was created by **Timothy Berners Lee** in 1989 at **CERN** in **Geneva**. World Wide Web came into existence as a proposal by him, to allow researchers to work together effectively and efficiently at **CERN**. Eventually it became **World Wide Web**. The following diagram briefly defines evolution of World Wide Web:

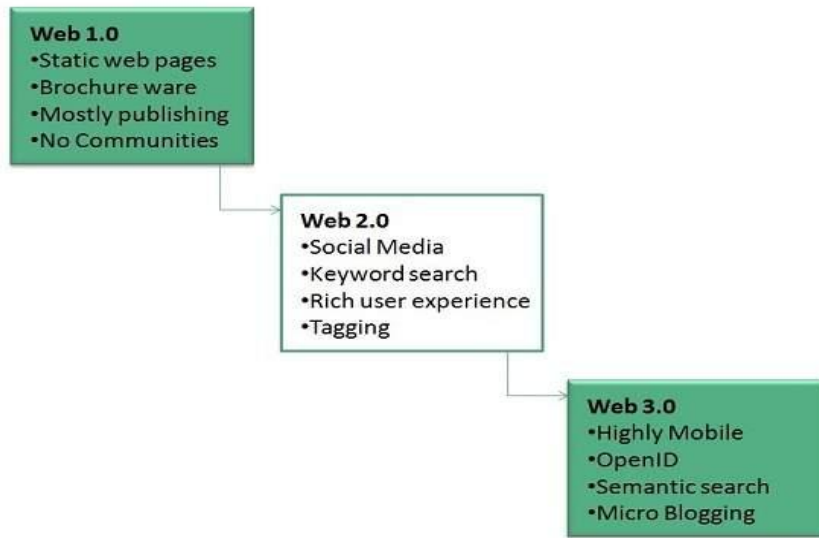


Figure 2-1: Evolution of WWW

2.3: Browsing the Web

A **Web browser**, or browser, is application software that allows users to access and view Web pages or access Web 2.0 programs. To browse the Web, you need a computer or mobile device that is connected to the Internet and has a Web browser. The more widely used Web browsers for personal computers are **Internet Explorer, Firefox, Opera, Safari, and Google Chrome**. With an Internet connection established, you start a Web browser. The browser retrieves and displays a starting Web page, sometimes called the **browser's home page** (Figure 2-2). The initial home page that is displayed is one selected by your Web browser. You can change your browser's home page at anytime. Another use of the term, **home page**, refers to the first page that a Web site displays and it provides information about the Web site's purpose and content. The home page usually contains links to other documents, Web pages, or Web sites. A link, short for hyperlink, is a built-in connection to another related Web page or part of a Web page.

For a computer or mobile device to display a Web page, the page must be downloaded. **Downloading** is the process of a computer or device receiving information, such as a Web page, from a server on the Internet. While a browser downloads a Web page, it typically displays an animated logo or icon in the browser window. The animation stops when the download is complete. The time required to download a Web page varies depending on the speed of your Internet connection and the amount of graphics involved.

2.3.1: How a Web Browser Displays a Home Page

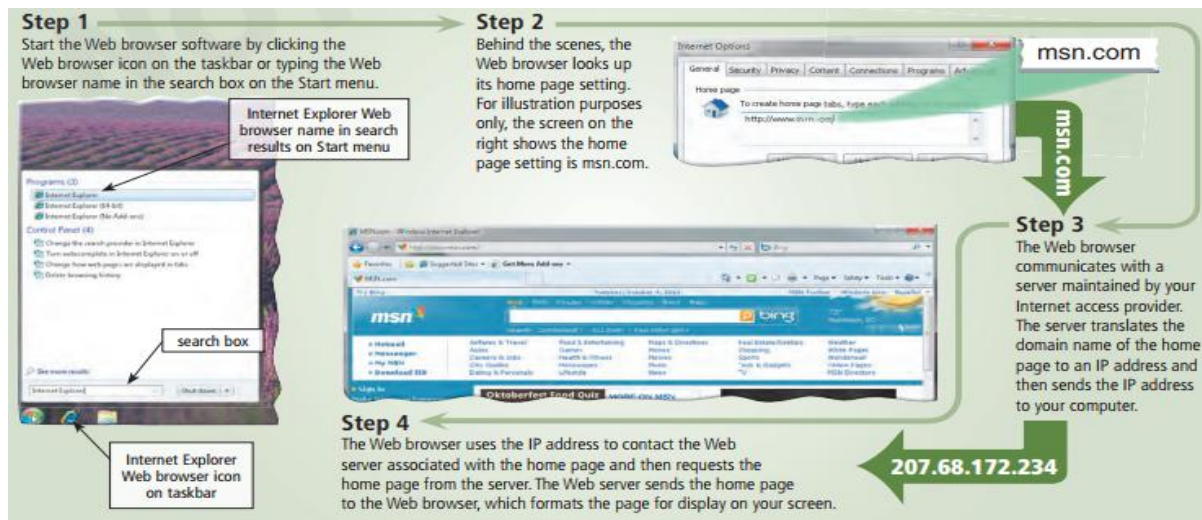


Figure 2-2: This figure shows how a Web browser displays a home page.

2.3.2: Internet Addresses

The Internet relies on an addressing system much like the postal service to send data and information to a computer at a specific destination. An **IP address**, short for Internet Protocol address, is a number that uniquely identifies each computer or device connected to the Internet. The IP address usually consists of four groups of numbers, each separated by a period. The number in each group is between 0 and 255. For example, the numbers 72.14.207.99 are an IP address. In general, the first portion of each IP address identifies the network and the last portion identifies the specific computer. These all-numeric IP addresses are difficult to remember and use. Thus, the Internet supports the use of a text name that represents one or more IP addresses. A **domain name** is the text version of an IP address. Figure 2-3 shows an IP address and its associated domain name. As with an IP address, the components of a domain name are separated by periods. The text in the domain name up to the first period identifies the type of Internet server. In Figure 1-8, for example, the www indicates a Web server.



Figure 2-3: The IP address and domain name for the Google Web site.

Every domain name contains a top-level domain (TLD), which is the last section of the domain name. A generic TLD (gTLD), such as the com in Figure 2-3 on the previous page, identifies the type of organization associated with the domain. The organization that assigns and controls top-level domains is the Internet Corporation for Assigned Names and Numbers (ICANN pronounced EYE-can). Figure 2-4 lists some gTLDs. For TLDs such as biz, com, info, name, net, and org, you register for a domain name from a registrar, which is an organization that sells and manages domain names. For international Web sites outside the United States, the domain name also includes a country code TLD (ccTLD), which is a two-letter country code, such as au for Australia. For example, www.philips.com.au is the domain name for Philips Australia. Some smaller countries have granted use of their ccTLDs for commercial purposes, such as tv (Tuvalu) for the television/entertainment industry. The **domain name system (DNS)** is the method that the Internet uses to store domain names and their corresponding IP addresses. When you specify a domain name, a DNS server translates the domain name to its associated IP address so that data and information can be routed to the correct computer. A **DNS server** is an Internet server that usually is associated with an Internet access provider.

Examples of Generic Top-Level Domains	
Generic TLD	Intended Purpose
aero	Aviation community members
biz	Businesses of all sizes
cat	Catalan cultural community
com	Commercial organizations, businesses, and companies
coop	Business cooperatives such as credit unions and rural electric co-ops
edu	Educational institutions
gov	Government agencies
info	Business organizations or individuals providing general information
jobs	Employment or human resource businesses
mil	Military organizations
mobi	Delivery and management of mobile Internet services
museum	Accredited museums
name	Individuals or families
net	Network providers or commercial companies
org	Nonprofit organizations
pro	Certified professionals such as doctors, lawyers, and accountants
tel	Internet communications
travel	Travel industry

Figure 2-4: Examples of Generic Top-Level Domains

2.3.3: Web Addresses

A Web page has a unique address, called a **URL (Uniform Resource Locator)** or **Web address**. For example, the home page for the United States National Park Service Web site has <http://www.nps.gov> as its Web address. A Web browser retrieves a Web page using its Web address. If you know the Web address of a Web page, you can type it in the Address bar at the top of the browser window. For example, if you type the Web address <http://www.nps.gov/grsm/planyourvisit/wildlifeviewing.html> in the Address bar and then press the enter key, the browser downloads and displays the Web page shown in Figure 2-5.

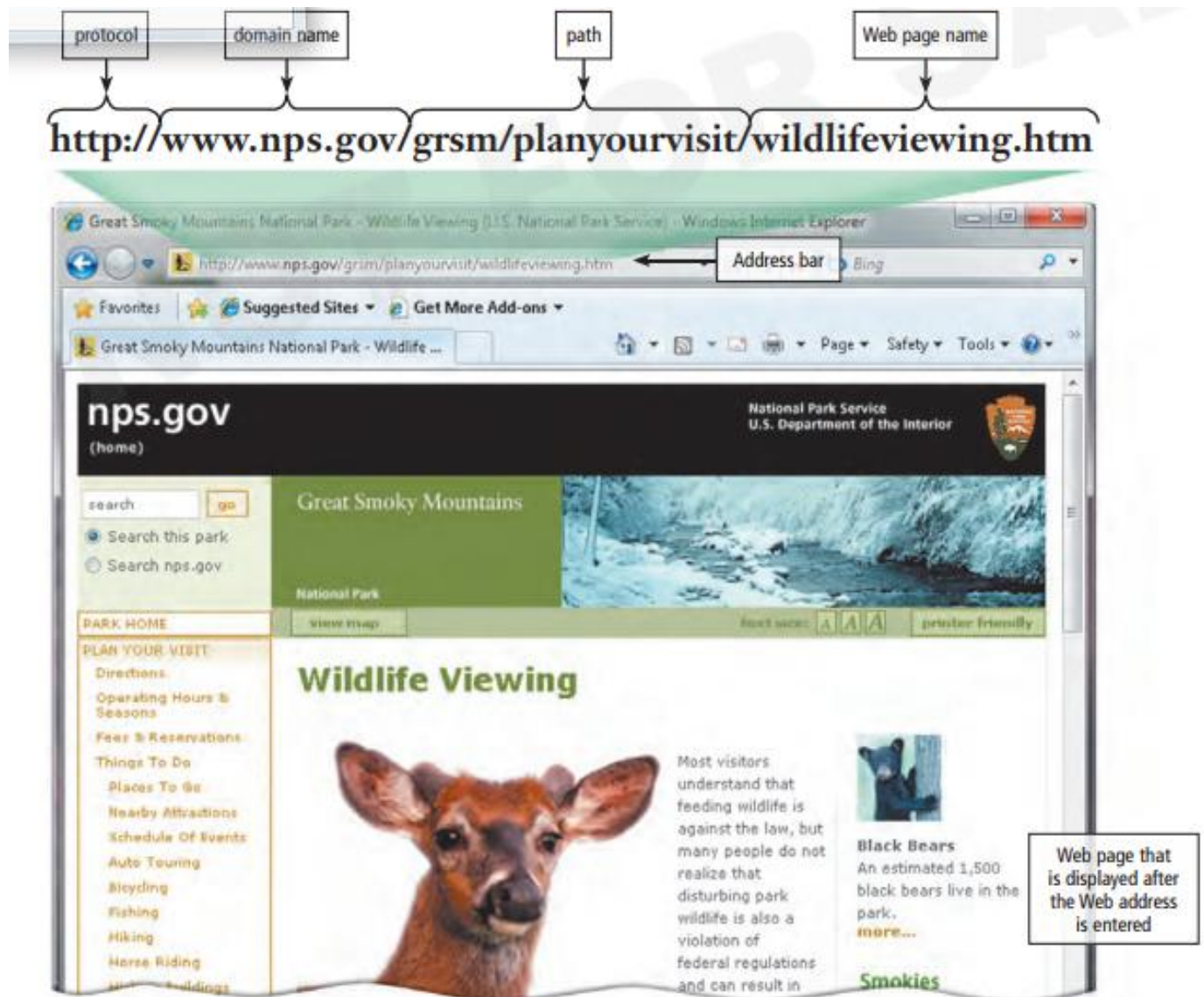


Figure 2-5: Web Page display

A URL is used to uniquely identify a resource over the web. URL has the following syntax:

Protocol: //hostname: port/path-and-file-name

There are 4 parts in a URL:

1. **Protocol:** The application-level protocol used by the client and server, e.g., HTTP, FTP, and telnet.
2. **Hostname:** The DNS domain name (e.g., `www.nowhere123.com`) or IP address (e.g., `192.128.1.2`) of the server.

3. **Port:** The TCP port number that the server is listening for incoming requests from the clients.
4. **Path-and-file-name:** The name and location of the requested resource, under the server document base directory.

For example, in the URL <http://www.nowhere123.com/docs/index.html>, the communication protocol is **HTTP**; the hostname is **www.nowhere123.com**. The port number was not specified in the URL, and takes on the default number, which is TCP port **80** for HTTP. The path and file name for the resource to be located is **"/docs/index.html"**.

2.3.4 Protocols governing the web

A Protocol is a set of rules. Protocols allow two computers to communicate over media such as wireless or hardwired technologies. When computers communicate with each other, there needs to be a common set of rules and instruction that each computer follows. Protocols are the language of computers.

Some types of Protocols:

a. Hyper Text Transfer Protocols (HTTP)

HTTP (Hypertext Transfer Protocol) is perhaps the most popular application protocol used in the Internet (or The WEB).

- HTTP is an *asymmetric request-response client-server* protocol as illustrated. An HTTP client sends a request message to an HTTP server. The server, in turn, returns a response message. In other words, HTTP is a *pull protocol*, the client *pulls* information from the server (instead of server *pushes* information down to the client).

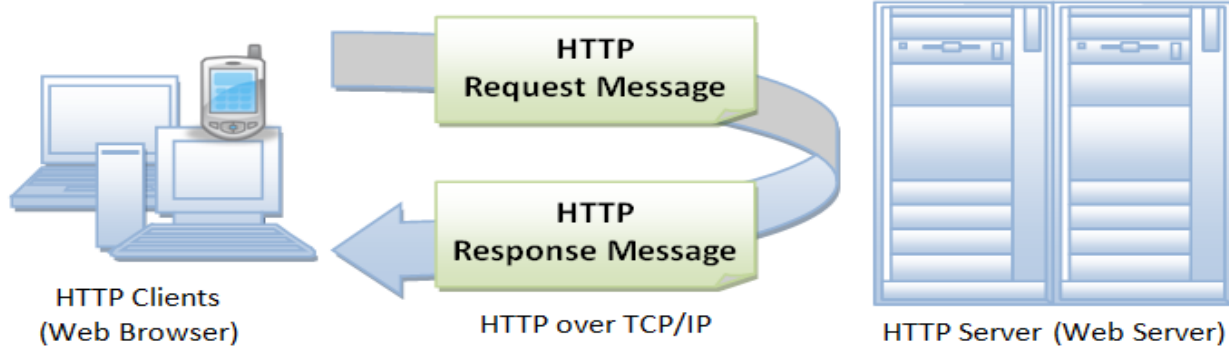


Figure 3-1: HTTP Protocol

- HTTP is a stateless protocol. In other words, the current request does not know what has been done in the previous requests.
- HTTP permits negotiating of data type and representation, so as to allow systems to be built independently of the data being transferred.
- Quoting from the RFC2616: "The Hypertext Transfer Protocol (HTTP) is an application-level protocol for distributed, collaborative, hypermedia information systems. It is a generic, stateless, protocol which can be used for many tasks beyond its use for hypertext, such as name servers and distributed object management systems, through extension of its request methods, error codes and headers."

Whenever you issue a URL from your browser to get a web resource using HTTP, e.g. <http://www.nowhere123.com/index.html>, the browser turns the URL into a *request message* and sends it to the HTTP server. The HTTP server interprets the request message, and returns you an appropriate response message, which is either the resource you requested or an error message.

This process is illustrated below:

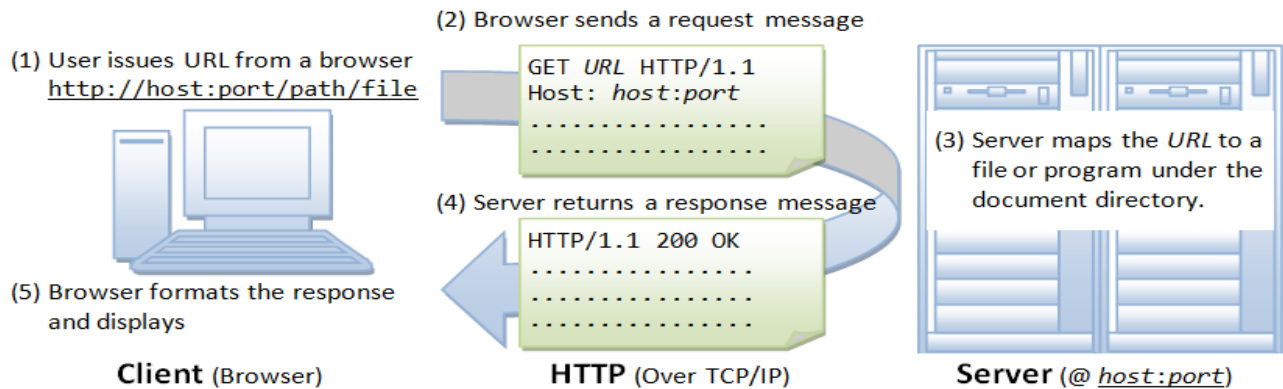


Figure 3-2: How HTTP works

b. Simple Mail Transfer Protocol (SMTP)

This protocol is used for transferring e-mail between computers. **Simple Mail Transfer Protocol**, and defines the method that handles the process of **email exchange and delivery** across IPs. To put it simply, when you send a message to a friend using a mail client, it's picked by an **outgoing server** (called **SMTP server**) which starts a conversation with your friend's incoming one.

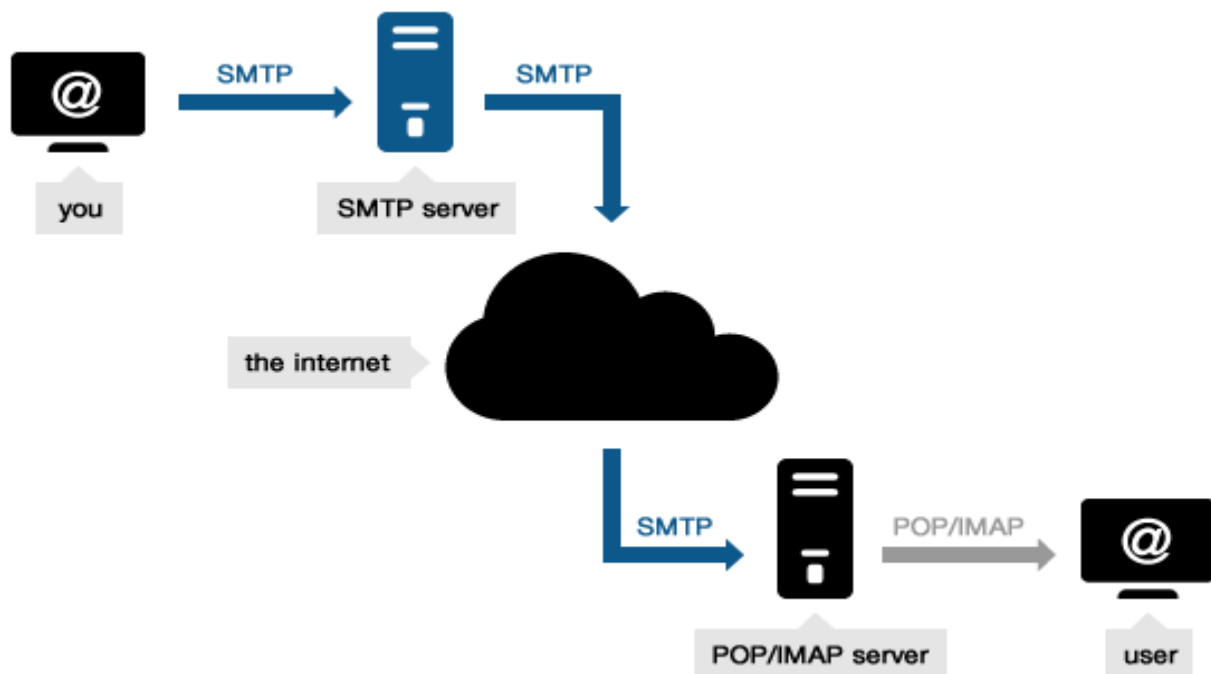


Figure 3-3: SMTP Protocol

SMTP and **POP3** are two of the most important technical aspects of the whole process of email dispatch and delivery. To put it simply, SMTP (acronym for **Simple Mail Transfer Protocol**)

handles the delivery, while POP (acronym for **Post Office Protocol**) takes care of the email retrieval and download. (Another very popular protocol that you can use in place of POP is **IMAP** – that is, Internet Message Access Protocol which allows the client program to manipulate the e-mail message on the server without downloading them on the local computer).

c. File Transfer Protocol (FTP)

FTP is used to upload files on server and download files from server. During this connection, the server and client are able to then transfer various files. This is done through converting files into binary and then transmitting the data, this concise process allows someone to download or upload a file to the internet.

FTP STEPS

- | | |
|-------------|--------------|
| 1. Request | 3. Transfer |
| 2. Response | 4. Terminate |

FTP: REQUEST

- This request is initiated by the client
- Allowing the File Transfer Protocol to begin
- Sub-commands
 - store, delete, lookup, open and close
 - Allows client to interact with the server before they download or upload a file

FTP: RESPONSE

- Server will inform the client that it is ready to either receive or send a file
- The sub-commands used in this transaction are “ready to send” and “ready to receive”

FTP: TRANSFER

- The actual transaction will occur in the transfer step

FTP: TERMINATE

- Two types of terminate transactions
 - successful terminate
 - unsuccessful terminate

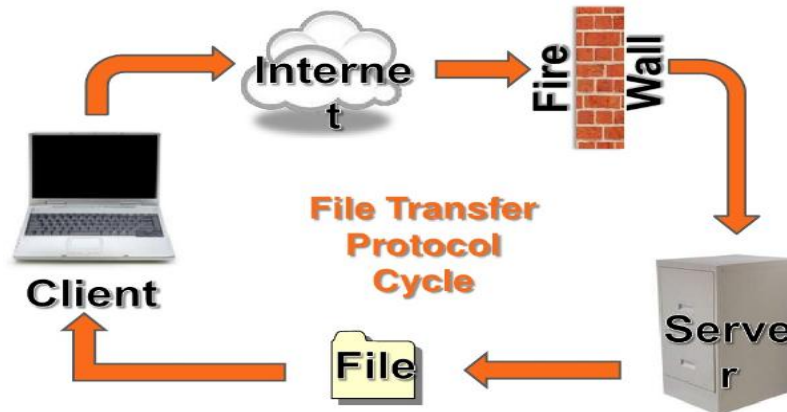


Figure 3-4: FTP Protocol

d. Transmission Control Protocol/Internet Protocol (TCP/IP)

TCP/IP stands for **Transmission Control Protocol/Internet Protocol**. It is the communication language or protocol used to connect hosts on the internet. TCP/IP works on client-server communication model; here the computer user as a client sends requests by another computer or server in the network. In TCP/IP, each communications start from one point (host computer) in the network to another point (another host computer).

How TCP/IP Works

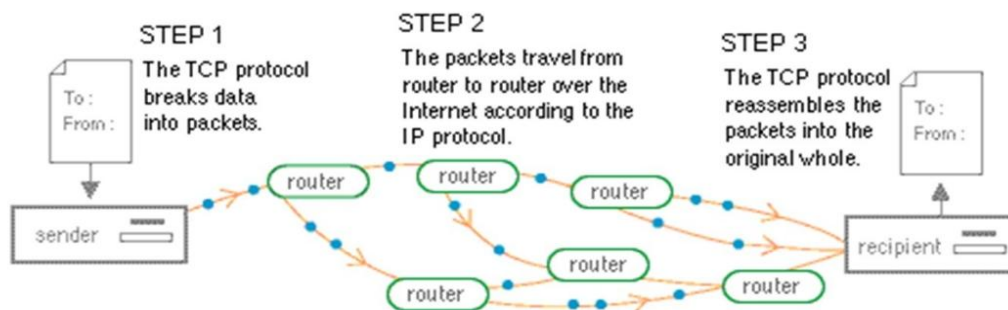


Figure 3-5: TCP/IP Protocol

2.3.5. Search Engines

The Web is a worldwide resource of information. A primary reason that people use the Web is to search for specific information, including text, pictures, music, and video. The first step in successful searching is to identify the main idea or concept in the topic about which you are seeking information. Determine any synonyms, alternate spellings, or variant word forms for the topic. Then, use a search tool to locate the information. Two types of search tools are search

engines and subject directories. A search engine is a program that finds Web sites, Web pages, images, videos, news, maps, and other information related to a specific topic. A subject directory classifies Web pages in an organized set of categories, such as sports or shopping, and related subcategories. Some Web sites offer the functionality of both a search engine and a subject directory. The table in Figure 4-1 lists the Web addresses of several popular general-purpose search engines and subject directories. A **search engine** is helpful in locating information for which you do not know an exact Web address or are not seeking a particular Web site. Thousands of search engines are available. Some search through Web pages for all types of information. Other search engines can restrict their searches to a specific type of information, such as the following items:

- Images — pictures, diagrams, and drawings.
- Videos — home videos, music videos, television programs, and movie clips.
- Audio — music, songs, recordings, and sounds.
- Publications — news articles, journals, and books.
- Maps — maps of a business or address, or driving directions to a destination.
- People or Businesses — addresses and telephone numbers.
- Blogs—specific opinions and ideas of others

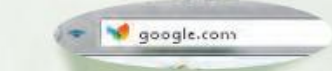
Search engines require that you enter a word or phrase, called **search text or search query**, which describes the item you want to find. Each word in the search text is known as a **keyword**. Your search text can be broad, such as spring break destinations, or more specific, such as Walt Disney World. Search engines often respond with thousands of results, whose content varies depending on the type of information you are seeking. Some results are links to Web pages or articles; other results are media, such as images or videos. You may find that many items that appear in the search results have little or no bearing on the item you are seeking. You can eliminate the superfluous items in your search results by carefully crafting search text that limits the search. If you misspell search text, many search engines identify alternative search text. Some also provide suggested keywords, links, and/or images as you begin typing your search text.

Widely Used Search Tools			
Search Tool	Web Address	Search Engine	Subject Directory
A9	a9.com	X	
AlltheWeb	alltheweb.com	X	
AltaVista	altavista.com	X	
AOL Search	search.aol.com	X	
Ask	ask.com	X	
Bing	bing.com	X	
Cuil (pronounced cool)	cuil.com	X	
Dogpile	dogpile.com	X	
Excite	excite.com	X	X
Gigablast	gigablast.com	X	X
Google	google.com	X	X
Lycos	lycos.com	X	
MSN	msn.com	X	X
Open Directory Project	dmoz.org	X	X
WebCrawler	webcrawler.com	X	
Yahoo!	yahoo.com	X	X

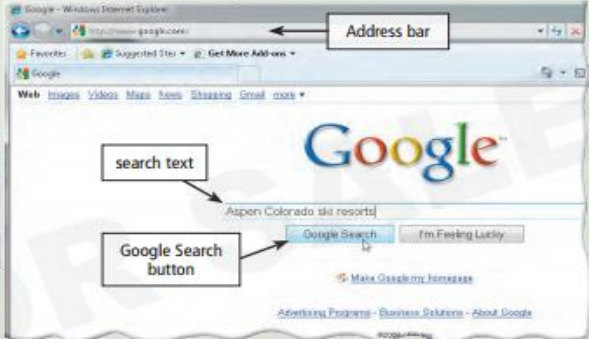
Figure 4-1: Popular search engines and subject directories.

How to Use a Search Engine

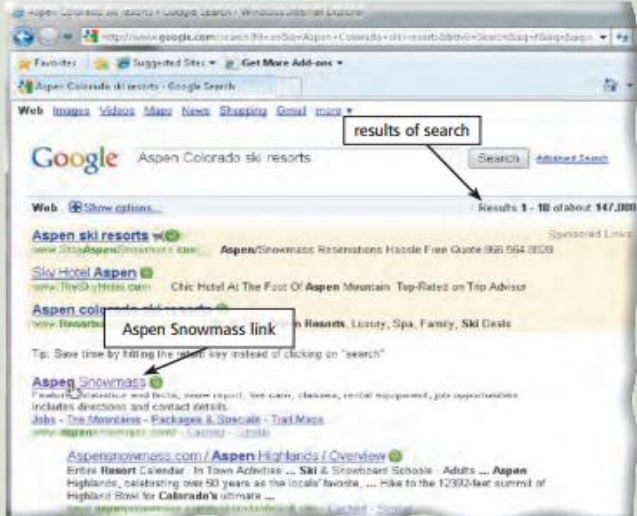
Step 1
Type the search engine's Web address (in this case, google.com) in the Address bar in the Web browser.



Step 2
Press the ENTER key. When the Google home page is displayed, type Aspen Colorado ski resorts as the search text and then point to the Google Search button.



Step 3
Click the Google Search button. When the results of the search are displayed, scroll through the links and read the descriptions. Point to the Aspen Snowmass link.



Step 4
Click the Aspen Snowmass link to display a Web page with a description and links to skiing in Aspen.

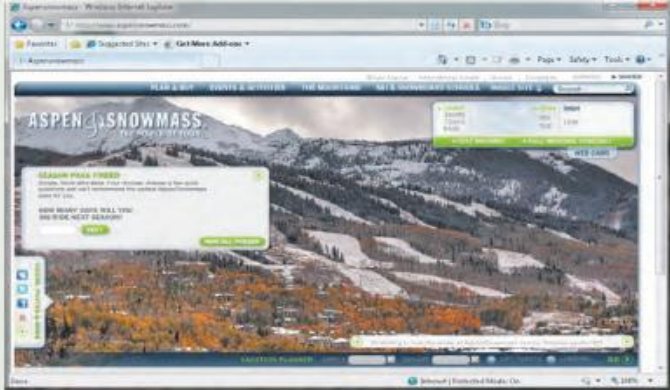


Figure 4-2: How to Use a Search Engine

2.4. E-mail Services

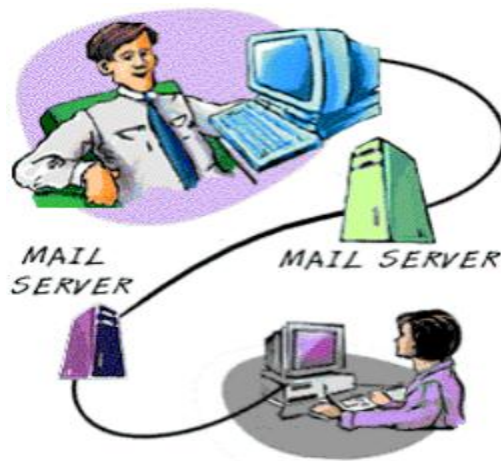
Electronic mail, or e-mail, is the most frequently used service on the Internet. E-mail Protocols are set of rules that help the client to properly transmit the information to or from the mail server.

Why use Email?

- ✓ You can send a message any time, anywhere.
- ✓ You can send the same message to several people at the same time.
- ✓ You can forward information to co-workers without retyping it.
- ✓ You can save time. E-mail is fast, usually taking no more than a few minutes to be received.
- ✓ You can e-mail electronic documents and the recipients can then edit and return revised versions.
- ✓ You can send messages around the world as easily as to co-workers in the next office.

How Email works on the Internet?

Like a postal address, an e-mail address specifies the destination of an electronic message. An Internet e-mail address looks like this: **user_name@domain_name**. The user name is a unique name that identifies the recipient. The domain name is the address. Many people can share the same domain name.



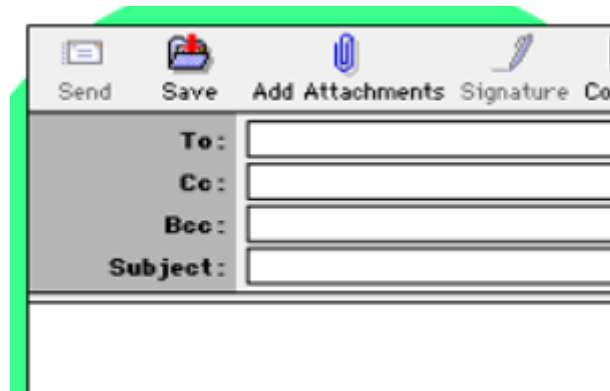
- E-mail is sent and received through electronic "post offices" known as mail servers.
- To read your e-mail, you must retrieve it from the mail server.
- Once you enter the address of the recipient, compose your message, and click Send, your e-mail software handles the delivery.

How reliable is Email?

- If the message doesn't reach its destination the first time, the mail server sends it again.
- If the message is not delivered, you usually receive a message explaining the problem, along with the full text of the original message. You can correct the problem usually an incorrect e-mail address--and resend it.

Anatomy of an Email message

- **To** contains the e-mail addresses of the **recipients**. This is a mandatory entry.
- **CC**, short for **Carbon Copy**. This is optional.
- **BCC**: short for **Blind Carbon Copy** contains the e-mail addresses of other recipients who receive copies, but their names and addresses are hidden from the other recipients. This is optional.
- **Subject** contains the **main topic** of the message. Keep this brief. Recipients see this in their summary of incoming e-mails.
- **Attachment** contains the **names of files** that you may be sending, for example, a word-processing document or a spreadsheet.



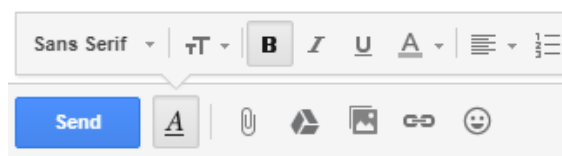
- Body contains the **message** itself.

Signatures

- E-mail can include a signature at the end of the message. An e-mail signature is not your hand-written signature. Like a business card, an e-mail signature can include your name, title, telephone and fax numbers, street address, and e-mail address.

- It saves time, you don't have to retype the information each time you send a message. Clicking the feature automatically adds it to the message.

Mr.Yenon W.Rinny
 Vice Rector/University of Washinton
 College of Science and Engineering
 Department of Communication Engineering
 +345-890-979056
 Skype ID: rinny1.yeno



Reading Email messages

- Messages are organized in folders or boxes. Received mail is kept in an Inbox; outgoing mail is stored in an Outbox; sent mail is filed in a Sent box.
- To read an e-mail, double-click on the envelope icon or the Subject line of the message. This opens a new window containing the header and body of the message.

Get Mail Delete To: Mail Re: Mail Re: All For		
Folder	Unread	Total
Inbox	1	48
Outbox	0	0
Sent Mail	0	177
Trash	5	80
Hot Air	0	0
More Hot Air	0	44
Reservations	0	0
Querries	0	5
Photographers	0	9
Fan Mail	0	4
Personal	0	16
Sent	0	0

Replying to Email messages

- ✓ The window with the e-mail you are reading has a Reply or Re button. Click on this button to open a new window. The **To** and **Subject** lines are automatically filled in with the e-mail address of the sender and the subject of the message.
- ✓ The text of the original e-mail appears in the body of the reply, usually marked in a different font or color, or with a special character, such as the greater than sign (>) at the beginning of each line.



2.5 Introduction to Web Design

2.5.1. Overview

We consider there are two types of web sites: **static web sites** and **dynamic web sites**.

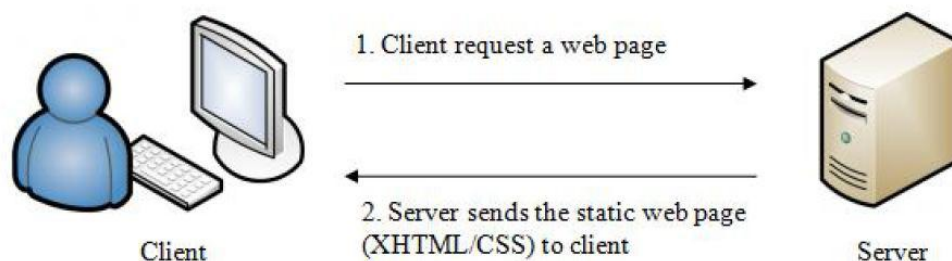
- **Static web sites:** web sites that are performed solely for the languages (X) HTML and CSS. They work very well but their contents cannot be updated automatically: it requires that the web site owner (webmaster) modifies the source code for adding new content. It's not very practical when you should update his web site several times in one day! Static web sites are well suited for making web sites "window" to present such a business, but without going further. This type of web site is increasingly rare today, because as soon as we add an element of interaction (like a contact form), we no longer speak of static web site but dynamic web site.
- **Dynamic web sites:** more complex, they use other languages in addition to (X) HTML and CSS, such as PHP and MySQL. The content of these web sites is called "**dynamic**" because it can change without the intervention of the webmaster! Most web sites you visit today are dynamic web sites. The only prerequisite for learning to create this type of web site is already known to make static web sites in XHTML and CSS.

2.5.2. How does a web site work?

a. For a static web site

When the web site is static, the pattern is very simple. This happens in two stages:

- * The client asks the server to see a web page.
- * The server responds by sending the requested page.



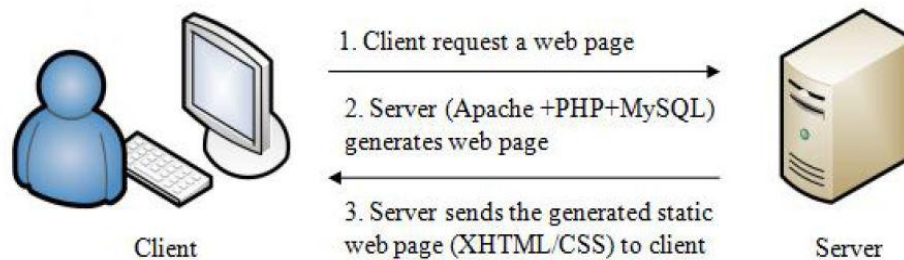
In the case of a static web site, the server stores web pages and sends back the entire web page

(Without any change in its initial content) to the client who is requesting the page them without Changing them.

b. For a dynamic web site with PHP and MySQL

When the web site is dynamic, there is an intermediate step: the page is generated.

- * The client asks the server to see a web page.
- * The server prepares the page specifically for the client's request.
- * The server sends the page it has generated.



The web page is generated each time a client (web browser) demands it. This is precisely what makes dynamic web sites alive: the content of a page can change from moment to moment.

This is how some web sites manage to display some information such as your username on all pages. Since the server generates a page every time the client requests it, the can customize it to suit the tastes and preferences of the client (such as his real name).

2.5.3: Programming languages for web sites.

1) For a **static web site**: XHTML and CSS

Many languages have been created to produce web sites. Two of them are an essential basis for all webmasters:

- ✓ **XHTML** is the language at the base of web sites. It is very similar to HTML but imposes some rules a little stricter. Wherever possible, I recommend using XHTML rather than HTML because it forces you to treat the source code of your web site. XHTML is a simple language to learn who operates out of tags.

An example of XHTML code:

<p>This is a snapshot of a paragraph of text!</p>

- ✓ **CSS** is the language formatting of web sites. While XHTML is used to write the contents of your web pages and the structure, the CSS handles the formatting and layout. This CSS is that we choose the particular color, size, menus and much more.

Here is a CSS code:

```
div.banner {
text-align: center;
font-weight: bold;
font-size: 120%;
}
```

2) For a dynamic web site: We add **PHP and MySQL**

What programs do we need?

Depending on whether one creates a static web site or a dynamic web site, you need different Software. In fact, making a dynamic web site for us unfortunately requires a bit more software!

With a static web site

The webmasters who create static web sites with XHTML and CSS are lucky; they usually already have all the programs they need:

- **A text editor:** in theory a program such as Notepad shipped with Windows only, although it is Recommended to use a bit more advanced tool like Notepad ++.
- **A web browser:** it can test the web page. You can use eg Mozilla Firefox, Internet Explorer, Google Chrome, Opera, Safari, or any other browser you are accustomed to go on the web. It is advisable to regularly test your web site on different browsers.
- **A web server** (such as Apache): This is responsible for hosting and issuing web pages to visitors. However, Apache only handles static web sites (it can only handle HTML pages). However, for those who work on dynamic web sites, these tools are not enough. It is necessary to install additional programs.